Scale bounds and force-class obstructions to physically realizing OpenAI's forced Navier–Stokes blowup

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Abstract

OpenAI's preprint constructs a smooth, compactly supported force $\boldsymbol{f}$ on $\mathbb{R}^3 \times (0,\infty)$ for which the Navier–Stokes equations with zero initial data have a solution that blows up at time 1. Taking the construction as correct, we show that no physical source we examine can supply the force it needs. First, by the paper's own constants, the swirling profile that the solution approaches extends to more than 7.20×10^{47} core lengths: for a core no smaller than the molecular spacing of water, the profile is more than 10^{12} times the Hubble radius, and in water the axis pressure falls below the vapor pressure for core lengths under 9.539 mm. Second, under two hypotheses of Constantin, Ignatova and Vicol that the OpenAI solution satisfies, no gradient force can produce such a singularity, and no force bounded in C^2 up to blowup can if it is a Coriolis force about the symmetry axis or its curl is real-analytic, quasianalytic or axisymmetric; nor can viscous Boussinesq buoyancy on $\mathbb{T}^3$ or the Lorentz force of axisymmetric toroidal magnetohydrodynamics on $\mathbb{R}^3$, given only a bound on the temperature or on B_θ/r. Third, for an admissible choice of its stage cutoffs, the OpenAI force lies in no Gevrey class with time-uniform constants. The exact core axisymmetry used here can be relaxed to double-exponential smallness under analytic bounds, which is sharp for criteria using only those bounds and the core size. None of this bears on the truth of the theorem or proves global regularity.

Keywords: Navier–Stokes equations, finite-time blowup, external forcing, axisymmetric flow, Gevrey class, unique continuation

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1 Introduction

The Clay problem statement [1] poses the Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u - \nu\Delta u + \nabla p = f, \qquad \nabla \cdot u = 0, \tag{1}$$

with viscosity $\nu > 0$, and calls f "a given, externally applied force (e.g. gravity)". "To give reasonable leeway to solvers while retaining the heart of the problem", it asks for a proof of one of four statements. Statements (C) and (D) ask for blowup on $\mathbb{R}^3$ or on $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$, and they ask of the force only smoothness and decay: condition (5) of [1] on $\mathbb{R}^3$, or condition (9) on $\mathbb{T}^3$. The problem statement motivates condition (5) with the words "For physically reasonable solutions, we want to make sure $u(x,t)$ does not grow large as $|x| \to \infty$".

OpenAI [2] constructs, for every $\nu > 0$, a force $f \in C_c^\infty(\mathbb{R}^3 \times (0,\infty); \mathbb{R}^3)$ and a smooth solution of (1) on $\mathbb{R}^3 \times [0,1)$ with $u(\cdot,0) = 0$, support in a fixed compact set, bounded energy, and $\limsup_{t\uparrow 1} \|u(t)\|_{L^\infty} = \infty$. The force is defined as the residual of a prescribed flow: "For any incompressible flow u and pressure p, we can always define the external force f to be the residual in (1.1)" [2]. The paper states that this establishes (C), and through compact support also (D). The terms of the prize are for the Clay Institute to apply, and we do not address them.

This paper compares the force with the forces of physical sources. The flow is a swirling core that collapses self-similarly, held together by the Reynolds stress of oscillatory pulses in dyadic bands outside the core; the force supplies whatever residual remains. Earlier forced constructions place the remainder of a layered construction in a force of limited regularity, $C^{1,1/2-\epsilon} \cap L^2$ for the three-dimensional Euler equations [3], or in a compactly supported smooth source for the two-dimensional incompressible porous media equation [4]; Alpöge and Buckmaster [5] use exact waves that do not advect themselves, with smooth forces in both equations of the two-dimensional inviscid Boussinesq system. OpenAI's force is smooth with compact support and vanishes to all orders at the singular point [2].

Two remarks fix the status of the construction. We checked the paper's text line by line and recomputed the exponent chains of its Lemmas 9.8 and 10.5 and its Appendix A, and found no step that fails; the gaps we found are in the prose (for example, the rates in step 3 of the proof of its Theorem 4.6 are used but not argued), and none affects the result. We built the Lean statement of (C) and (D) in OpenAI's repository [2], and its axioms are only the three standard ones of Lean's library; Bastounis, Circelli and Hansen [6] show that the intermediate Lean lemmas do not follow the text, so the formal proof certifies the theorem but not the paper's argument. Cao, Chi and Nie [7] show that forces producing blowup from zero data are dense in $L^1_t H^s_x$ exactly when $s < 1/2$, so blowup by forcing as such is not the issue here; the issue is the class of the force.

The paper first fixes the setting and the notation of the construction. It then computes, from the construction's constants, the size of the leading field and what this means for water. It then proves that the physical body forces we consider cannot produce a singularity of the kind the construction builds, places the OpenAI force, for an admissible choice of its stage cutoffs, outside every Gevrey class with constants uniform in time, and determines how far the exact axisymmetry of the core used in those proofs can be weakened.

2 Setting

We use the notation of [2], which sets $\nu = 1$. The blowup time is $t = 1$; put $\tau = 1 - t$. With $0 < h < 1/100$, $A = 1/2 + h$ and $D = 1/2 - h$, the similarity coordinates (q, η, X) are defined, in cylindrical coordinates (r, θ, z) about the symmetry axis with unit vectors e_r, e_θ, e_z, by

$$\tau = q(1 - \eta^2), \qquad z = q^D \eta, \qquad X = \frac{r^2}{2q}, \tag{2}$$

with $q > 0$ and $-1 < \eta < 1$ [2, (3.2)]. On the plane $\eta = 0$ we have $q = \tau$. The leading field $u^{(0)}$ is axisymmetric, with azimuthal component and pressure

$$u^{(0)}_\theta = q^{-A} E(X, \eta), \qquad p^{(0)} = q^{-2A} \Pi(X, \eta), \tag{3}$$

where E and Π are fixed profiles [2, (4.3)]. The residual $\mathcal{R}(u, p) = \partial_t u + (u \cdot \nabla)u - \Delta u + \nabla p$ of the constructed flow satisfies $|\partial_x^\alpha \partial_t^b \mathcal{R}(u, p)| \le C_{\alpha,b,N,X_1} q^N$ on $0 \le X \le X_1$ as $q \downarrow 0$, for every spatial multi-index α, all nonnegative integers b and N, and every finite X_1, and it vanishes for $X \ge X_{\mathrm{ext}}$, a constant of the construction [2, (3.4)]. The force is this residual after a final spatial localization [2, Proposition 10.1].

The construction fixes its parameters in a stated order [2, Lemma 4.8, Proposition 4.10]. The parameters M_d and P_* are large, $T_d = e^{M_d} + 10$, $P_* > e^{T_d}$, and $0 < h < \min\{1/100, \lambda, e^{-T_d}\}$, where $\lambda > 0$ is the decay exponent of the profile in [2, (4.30)]. An amplitude C is chosen after P_* with $C^{-1} \ll P_*^{-1}$, so $C > P_*$. The reference radius and the matching radius are

$$X_R = 110(CP_*)^{10}, \qquad X_h = X_R e^{-5}. \tag{4}$$

Near X_h the profiles equal $E = P_* F(\eta) x^{1/10}$ with $x = X/X_R$ and $F(\eta) = (1 + \eta^2)^{-1}$, and the axis pressure is $\Pi(0, \eta) = \Pi_0(\eta)$ with

$$\Pi_0(\eta) \le -\tfrac{5}{2} P_*^2 F(\eta)^2, \tag{5}$$

where the pressure is normalized to vanish at radial infinity [2, Lemma 4.8(i), Proposition 4.10(iii)]. Since $M_d > 0$, $T_d > 11$, so

$$P_* > e^{11}, \qquad C > e^{11}, \qquad h < e^{-11} < 1.68 \times 10^{-5}. \tag{6}$$

The construction is a family of forces, not one force: its Lemma 5.4 states that stage cutoffs $a_{j+1} \ge 2a_j$ with the required properties exist, and each permitted sequence (a_j) gives a different field. Each statement about the force holds for every member of the family unless it names a choice of (a_j).

3 Physical scale of the construction

Fix a length unit L. Equation (1) with $\nu = 1$ becomes the physical equation with viscosity ν when positions are multiplied by L, times by L^2/ν, velocities by ν/L and kinematic pressures by ν^2/L^2. Define the *core length* $l_r = Lq^{1/2}$, the physical length that corresponds to $X = 1/2$ in (2). Then a point at similarity radius X lies at physical radius

$$r_{\mathrm{phys}} = l_r \sqrt{2X}, \tag{7}$$

and (3) gives the physical swirl speed and pressure of the leading field, for a fluid of density ρ:

$$u_\theta = \frac{\nu}{l_r} q^{-h} E, \qquad p = \rho \frac{\nu^2}{l_r^2} q^{-2h} \Pi. \tag{8}$$

Proposition 1 (Scale of the leading field)**:** *On the plane $\eta = 0$, at every time with $0 < q \le 1$, the leading field (3) has the following properties.*

(i) At $X = X_h$ its swirl speed satisfies $u_\theta \ge e^{10.5}\,\nu/l_r$.
(ii) On the axis its pressure, relative to the pressure at radial infinity, satisfies $p \le -\frac{5}{2}e^{22}\rho\nu^2/l_r^2$.
(iii) The matching radius $r_h = l_r\sqrt{2X_h}$ satisfies $r_h > \sqrt{220}\,e^{107.5}\,l_r > 7.20 \times 10^{47}\,l_r$.
(iv) The Reynolds number $u_\theta r_h/\nu$ at $X = X_h$ exceeds $\sqrt{220}\,e^{118} > 2.61 \times 10^{52}$.

Proof On $\eta = 0$, $q = \tau \le 1$, so $q^{-h} \ge 1$ and $q^{-2h} \ge 1$. At $X = X_h$, $x = e^{-5}$ and $F(0) = 1$, so $E = P_* e^{-1/2}$, and (6) gives $E > e^{10.5}$; (8) gives (i). By (5), $\Pi(0,0) \le -\frac{5}{2}P_*^2 < -\frac{5}{2}e^{22}$, and (8) gives (ii). By (4) and (6), $CP_* > e^{22}$, so $2X_h > 220e^{220}e^{-5}$, and (7) gives (iii). The product of (i) and (iii), divided by ν, is (iv). □

Proposition 1 concerns the leading field, which the solution approaches as $q \downarrow 0$. The bound on the approach is weak: on the circle of [2, (3.6)] the swirl is $\tau^{-A}(e_0 + O(\tau^{2h}))$ with a constant $e_0 > 0$, and by (6)

$$\tau^{2h} > 0.9976 \quad \text{for } \tau \ge 10^{-30}. \tag{9}$$

So the error bound of [2, (3.6)] is not small relative to the leading term at any time a laboratory could resolve, and it does not show that the leading field describes the flow at such times. The factor q^{-h} in (8) that makes the solution Type II also stays near one: $q^{-h} < 1.00116$ for $q \ge 10^{-30}$.

The ratio r_h/l_r is a pure number, the same in every fluid and at every time. A leading field with core length l_r therefore has matching radius more than $7.20 \times 10^{47}\,l_r$. Table 1 evaluates this for water. The Hubble radius $R_H = c/H_0 = 1.3733 \times 10^{26}$ m, with $H_0 = 67.36$ km s^{-1} Mpc^{-1} [8] and c the speed of light [9], contains r_h only if

$$l_r \le \frac{R_H}{7.20888 \times 10^{47}} = 1.90503 \times 10^{-22}\ \text{m}, \tag{10}$$

which is $1/(4.41 \times 10^6)$ of the proton charge radius $r_p = 8.4075 \times 10^{-16}$ m [9]. At that core length, Proposition 1 gives a swirl speed above 1.91×10^{20} m/s and an axis pressure deficit above 2.48×10^{44} Pa in water.

For water at 293.15 K and 0.101325 MPa we use the density $\rho = 998.207$ kg m^{-3}, dynamic viscosity 1.00160×10^{-3} Pa s (so $\nu = 1.00340 \times 10^{-6}$ m^2 s^{-1}), sound speed 1482.35 m/s and saturation pressure 2339.32 Pa [10], and the molecular spacing $(M/(\rho N_A))^{1/3} = 3.106 \times 10^{-10}$ m with $M = 18.0153$ g/mol [10] and the Avogadro constant N_A [9]. The axis pressure of the leading field is below the saturation pressure once its deficit exceeds the ambient pressure less the saturation pressure, 98985.68 Pa. Below the saturation pressure water can stay liquid in a metastable state until vapor bubbles nucleate and cavitation starts [11].

Table 1 Thresholds for the leading field in water at 293.15 K and 1 atm. Each row gives the core length below which the event occurs, by Proposition 1, and the matching radius r_h at that core length, in meters and in units of the Hubble radius R_H.

Event	**Condition on l_r**	**r_h (m)**	**r_h/R_H**
axis pressure below vapor pressure	$l_r < 9.539 \times 10^{-3}$ m	$> 6.876 \times 10^{45}$	$> 5.00 \times 10^{19}$
swirl speed at X_h above sound speed	$l_r < 2.458 \times 10^{-5}$ m	$> 1.772 \times 10^{43}$	$> 1.29 \times 10^{17}$
core length equal to molecular spacing	$l_r = 3.106 \times 10^{-10}$ m	$> 2.239 \times 10^{38}$	$> 1.63 \times 10^{12}$
swirl speed at X_h above speed of light	$l_r < 1.215 \times 10^{-10}$ m	$> 8.762 \times 10^{37}$	$> 6.38 \times 10^{11}$

The table gives a dichotomy for the leading field in water. Either its core length is at least the molecular spacing, and its matching radius is more than 10^{12} times the Hubble radius, or its core is smaller than a molecule, where the continuum equations (1) do not describe the fluid. The second branch holds for every blowup scenario near the blowup time; the first is specific to this construction, and it describes the solution at a given time only as accurately as the leading field approximates the solution then.

4 Physical body forces

4.1 The class of singularities

Constantin, Ignatova and Vicol (CIV) [12] isolate the structure of the OpenAI singularity. Let (u, π) be a suitable weak solution of (1) with $\nu = 1$ on the cylinder $Q = B(1) \times (-1, 0)$, where $B(r)$ is the open ball of

Table 2 Notation.

Symbol	Meaning
$\tau = 1 - t$	time to blowup, in the units of [2]
q, η, X	similarity coordinates (2)
h	Type II exponent, $h < e^{-11}$
E, Π	swirl and pressure profiles of the leading field (3)
P_*, C, M_d, T_d	parameters of the construction
X_R, X_h	reference and matching radii (4)
l_r	core length, $Lq^{1/2}$
r_h	physical radius at $X = X_h$
R_H	Hubble radius c/H_0

radius r about the origin, smooth on compact subsets of Q, with singular point $(0,0)$. Let Q_φ be the rotation through angle φ about the z axis, and define

$$(R_\varphi u)(x,t) = Q_\varphi^{-1} u(Q_\varphi x, t), \qquad \mathsf{P}u = \frac{1}{2\pi}\int_0^{2\pi} R_\varphi u \, d\varphi, \tag{11}$$

with $v = \mathsf{P}u$ and $w = u - v$ [12, (1.4)]. CIV's hypotheses are, with $0 < h < 1/2$:

(1.5) $\sup_{-1<t<0} \|f(\cdot,t)\|_{C^2(B(1))} < \infty$;
(1.6) f is real-analytic in x, locally uniformly on cylinders compactly contained in Q;
(1.7) $\|\partial_r^a \partial_z^b v_r(\cdot,t)\|_{L^\infty(B(1))} \le C_{\rm axi}(-t)^{-1/2-a/2-(1/2-h)b}$, and the same for v_z and v_θ with an extra factor $(-t)^{-h}$, for $a + b \le 2$;
(1.8) for each $t \in (-1,0)$ there is $\rho(t) > 0$ with $w(\cdot,t) = 0$ on $B(\rho(t))$.

CIV prove that (1.5) to (1.8) imply $u = v$ on Q and that $(0,0)$ is a regular point [12, Theorem 1.1]. They show that the OpenAI solution, translated and rescaled to Q, satisfies (1.7) and (1.8), and that its force does not satisfy (1.6) [12, Remark 1.2]. In (1.8) "No regularity or quantitative lower bound is assumed on ρ". We call a singularity *of the CIV class* if it satisfies (1.7) and (1.8).

4.2 Unique continuation of the rotation defect

Theorem 1 (Unique continuation of the rotation defect)**:** *Let (u,π) be a suitable weak solution of (1) on Q, smooth on compact subsets of Q, that satisfies (1.8). Suppose that the force satisfies*

(UC) for every $t \in (-1,0)$ and every φ, if $\operatorname{curl} f(\cdot,t) - R_\varphi \operatorname{curl} f(\cdot,t)$ vanishes on some ball $B(r)$, $r > 0$, then it vanishes on $B(1)$.

Then $u = v$ on Q. If moreover (1.5) and (1.7) hold, then $(0,0)$ is a regular point.

Proof Fix φ and put $U = u - R_\varphi u$. Rotation about the z axis maps $B(1)$ onto itself and commutes with (1), so $R_\varphi u$ solves (1) with force $R_\varphi f$, and

$$\partial_t U - \Delta U + (u\cdot\nabla)U + (U\cdot\nabla)R_\varphi u + \nabla \pi_\varphi = f - R_\varphi f, \qquad \nabla\cdot U = 0, \tag{12}$$

with $\pi_\varphi = \pi - \pi \circ Q_\varphi$. Since v is axisymmetric, (1.8) gives $R_\varphi u = v = u$ on $B(\rho(t))$, so $U(\cdot,t) = 0$ on $B(\rho(t))$.

Fix a compact interval $J \subset (-1,0)$. The sets $E_n = \{t \in J : U(\cdot,t) = 0 \text{ on } B(1/n)\}$ are closed, because U is continuous on compact subsets of Q, and their union is J. By the Baire category theorem some E_n contains an interval J'. On $B(1/n)\times J'$, (12) gives $f - R_\varphi f = \nabla \pi_\varphi$, so $\operatorname{curl} f - R_\varphi \operatorname{curl} f = 0$ there, since curl commutes with rotations. By (UC) this holds on $B(1)$ for each $t \in J'$, and since $B(1)$ is simply connected, $f - R_\varphi f = \nabla g_\varphi$ on $B(1)$. Then U solves on $B(r_0)\times J'$, for every $r_0 < 1$, a Stokes system of the form of [13, (1.3)] with bounded coefficients u and $R_\varphi u$, writing $(U\cdot\nabla)R_\varphi u = \nabla\cdot(U \otimes R_\varphi u)$, and zero force, the gradient being absorbed into the pressure, and U vanishes on the open set $B(1/n)\times J'$. Fabre's unique continuation theorem [13, Theorem 1.4] gives $U = 0$ on $B(r_0)\times J'$, hence on $B(1)\times J'$.

So the closed set $G = \{t \in (-1,0) : U(\cdot,t) = 0 \text{ on } B(1)\}$ meets every interval, and is therefore all of $(-1,0)$. As φ was arbitrary, $u = R_\varphi u$ for all φ, that is, $u = v$.

For the second statement, $u = v$ gives $f - R_\varphi f = \nabla \pi_\varphi$ on $B(1)$ for all φ, and averaging over φ (with π_φ chosen as a line integral from 0, which is smooth in φ) gives $f - \mathsf{P}f = \nabla g$. Replacing f by $\mathsf{P}f$ and π by $\pi - g$ leaves u unchanged; $\mathsf{P}f$ satisfies (1.5), and u is axisymmetric and satisfies (1.7). CIV's Theorem 1.3, which is the step by which they conclude Theorem 1.1 from $u = v$, gives regularity at $(0,0)$. □

Condition (UC) holds in three cases.

(a) $\operatorname{curl} f(\cdot,t)$ is real-analytic on $B(1)$ for each t, by the identity theorem. No uniformity in t is needed, unlike (1.6).
(b) For each t, the restriction of $\operatorname{curl} f(\cdot,t)$ to every line segment in $B(1)$ lies in one quasianalytic Denjoy–Carleman class C^M in the sense of Hörmander's Definition 1.3.7 [14]. The class is invariant under rotation and is a vector space, so $g = \operatorname{curl} f - R_\varphi \operatorname{curl} f$ is in it; if g vanishes on $B(r)$, its restriction to the segment from 0 to any $x \in B(1)$ vanishes to infinite order at 0, hence on the whole segment. The Denjoy–Carleman theorem [14, Theorem 1.3.8] characterizes such classes.
(c) $R_\varphi \operatorname{curl} f = \operatorname{curl} f$ for all φ, trivially.

Case (c) answers the question CIV pose in their Remark 2.5, "whether in Theorem 1.1 assumption (1.6) may be replaced by the rotation invariance (2.8) of $\operatorname{curl} f$": it may.

Corollary 1 (Forces with axisymmetric curl)**:** *On $B(1)$, $R_\varphi \operatorname{curl} f = \operatorname{curl} f$ for all φ holds if and only if $f = \mathsf{P}f + \nabla g$ for some function g. So a force whose non-axisymmetric part is a gradient cannot produce a singularity of the CIV class when it satisfies (1.5).*

Proof If $f = \mathsf{P}f + \nabla g$, then $R_\varphi \mathsf{P} f = \mathsf{P}f$ and $\operatorname{curl} R_\varphi \nabla g = \operatorname{curl} \nabla(g \circ Q_\varphi) = 0$. Conversely, $\operatorname{curl}(f - R_\varphi f) = 0$ on the simply connected ball gives $f - R_\varphi f = \nabla g_\varphi$ with $g_\varphi(x) = \int_0^1 (f - R_\varphi f)(sx) \cdot x \, ds$, which is smooth in φ; averaging over φ gives $f - \mathsf{P}f = \nabla g$ with g the average of g_φ. The second statement is Theorem 1 with case (c). □

4.3 Physical forces

Gradient forces. Uniform gravity and every force $f = \nabla g$ have $\operatorname{curl} f = 0$, so they satisfy (UC), and Theorem 1 applies. Directly, ∇g is absorbed into the pressure and u is the unforced solution.

Proposition 2 (Rotation about the symmetry axis)**:** *Let u solve (1) in a frame rotating with constant angular velocity Ωe_z, with the Coriolis force $-2\Omega e_z \times u$, the centrifugal force and a force f that satisfies (UC). If u satisfies (1.7) and (1.8), and f satisfies (1.5), then $(0,0)$ is a regular point.*

Proof The centrifugal force is the gradient $\nabla(\Omega^2 r^2/2)$. Pass to the inertial frame, $y = Q_{\Omega t} x$ and $u_{\text{in}}(y,t) = Q_{\Omega t}\big(u(x,t) + \Omega e_z \times x\big)$; then u_{in} solves (1) with the force $f_{\text{in}}(y,t) = Q_{\Omega t} f(Q_{\Omega t}^{-1} y, t)$, up to a gradient, and no Coriolis term. The bound $|u_{\text{in}}| \le |u| + |\Omega|\,|x|$ and the fact that the change of frame is a smooth isometry of Q onto itself make u_{in} a suitable weak solution on Q, smooth on compact subsets. Since $Q_{\Omega t}$ is a rotation about the z axis, it commutes with R_φ and P: the non-axisymmetric part of u_{in} is the rotated w, so (1.8) holds; f_{in} satisfies (UC) and (1.5); and the axisymmetric part changes by $\Omega r e_\theta$, whose derivatives are bounded, so (1.7) holds with a larger constant. Theorem 1 applies to u_{in}. □

A rotation axis tilted from the symmetry axis can destroy axisymmetry for nonzero symmetric data; Proposition 2 does not cover that case.

Buoyancy and the Lorentz force come from fields with their own evolution equations, coupled to u. For these closed systems the force is not prescribed but evolves with u.

Proposition 3 (Boussinesq and toroidal magnetohydrodynamics)**:** *Let (u,θ) be a smooth solution on $\mathbb{T}^3 \times [t_0, 0)$ of*

$$\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = \theta e_3, \qquad \partial_t \theta + u \cdot \nabla \theta = \kappa \Delta \theta, \qquad \nabla \cdot u = 0, \tag{13}$$

where $\nu, \kappa > 0$ and e_3 is the vertical unit vector. Assume that u and θ have mean zero, as in Cheng and Xu [15]. If u satisfies (1.7) and (1.8) after a neighborhood of a point $(x_0, 0)$ is translated and rescaled to Q with the axis of Q parallel to e_3, then $(x_0, 0)$ is a regular point. No bound on derivatives of θ is assumed. The same holds on $\mathbb{R}^3$ for magnetohydrodynamics with resistivity $\eta_{\text{m}} > 0$ when u and $B = B_\theta(r,z,t) e_\theta$ are axisymmetric about the z axis on $\mathbb{R}^3 \times [t_0, 0)$, the point x_0 lies on that axis, and u and B_θ / r are bounded on $\mathbb{R}^3 \times [t_0, t]$ for each $t < 0$.

Proof (i) By the maximum principle, $|\theta| \le \|\theta(t_0)\|_{L^\infty}$. (ii) Cheng and Xu [15, Theorem 2.1] prove that H^1 data give a unique solution that is spatially analytic for $0 < t < T_1$, where T_1 depends only on the H^1 norms of the data. Restarting from $(u(s), \theta(s))$ for $s < t$ close to t shows that $\theta(\cdot, t)$ is real-analytic for every $t < 0$. So $\operatorname{curl}(\theta e_3)$ satisfies (UC) by case (a), and Theorem 1 gives $u = v$. Then (12) with $U = 0$ gives $(\theta - \theta \circ Q_\varphi) e_3 = \nabla \pi_\varphi$, so $\theta - \theta \circ Q_\varphi$ has zero horizontal gradient; it vanishes on the axis, so θ is axisymmetric. (iii) It remains to run CIV's proof of their Theorem 1.3 with θ bounded but its derivatives unbounded. The force enters that proof as $(\operatorname{curl} f)_\theta$, as $g = (\operatorname{curl} f)_\theta / r$, and as $r f_\theta$ [12, Section 3]. Here $f_\theta = 0$, so the circulation and swirl sources vanish. By (i), $f = \theta e_3$ is bounded, so f lies in the parabolic Morrey space of the criterion of Gustafson, Kang and Tsai [16, Theorem 1.1], which does not require a divergence-free force. With f

itself in place of the divergence-free force of CIV's footnote 2, CIV's proof of their Lemma 3.2 gives an annulus on which u is bounded. On it the lemma uses a bound on f in C^2 in x, uniform up to $t = 0$ [12, Section 3]. To obtain this bound, put $\omega = \operatorname{curl} u$ and write $\partial_t\theta - \kappa\Delta\theta = -\nabla\cdot(u\theta)$ and $\partial_t\omega - \nu\Delta\omega = \operatorname{curl}(u\times\omega) + \operatorname{curl}(\theta e_3)$. Heat potentials over backward parabolic cylinders of a fixed radius, on shrinking annuli, bound in turn ω in L^∞, starting from $\nabla u \in L^2$, then θ in $C^{1/2}$, in $C^{1,1/2}$ and in $C^{2,1/2}$ in x. The last two steps use a bound on u in $C^{1/2}$ in x, which follows from the bounds on u and ω in L^∞ since $\nabla\cdot u = 0$. Each step treats the drift terms as sources and needs no regularity in t, and the bounds hold up to $t = 0$ because the cylinders lie in the past. In CIV's Section 4, let t_n, z_n and r_n be the times and the axial and radial centers of their contradiction sequence [12, (4.1)], and put $\theta_n(R, Z, T) = \theta((-t_n)^{1/2}R, z_n + (-t_n)^{1/2-h}Z, (-t_n)T)$. Here $(\operatorname{curl} f)_\theta = -\partial_r\theta$, so the source (4.8a) of CIV is $\mathcal{G}_n = -(-t_n)^{3/2+h}R^{-1}\partial_R\theta_n$. For $\mathcal{X} \in \mathbb{R}^4$ with $R = |\mathcal{X}|$, $R^{-1}\partial_R\theta_n = \nabla_{\mathcal{X}}\cdot(\theta_n\mathcal{X}/R^2) - 2\theta_n/R^2$. On compact sets $\theta_n\mathcal{X}/R^2$ is bounded in L^p for $p < 4$ and θ_n/R^2 in L^p for $p < 2$, so $\mathcal{G}_n \to 0$ in $W^{-1,p}_{\mathrm{loc}}$ for $1 < p < 2$. The source (4.8b) of CIV is $-(-t_n)^{3/2+h}$ times the R-derivative of θ at their point (4.17), with θ regarded as a function of R through $r = r_n + (-t_n)^{1/2}R$, and it tends to zero in $W^{-1,\infty}_{\mathrm{loc}}$. These bounds give the time compactness that CIV use away from the axis and remove the sources from the limits. Let ζ be CIV's smooth interior cutoff. Put $Y = \|\zeta\omega\|_2^2$. In the enstrophy inequality [12, (5.4)], the force term is $\int\zeta^2\omega\cdot\operatorname{curl} f = \int f\cdot\operatorname{curl}(\zeta^2\omega)$, and Young's inequality bounds it by $\frac{1}{4}\|\zeta\nabla\omega\|_2^2 + C(Y+1)$, where C depends on $\|f\|_\infty$. The final step uses the same criterion. For toroidal magnetohydrodynamics on $\mathbb{R}^3$, u is axisymmetric, so $u = v$ and step (ii) is not needed. The function $\Psi = B_\theta/r$ satisfies $\partial_t\Psi + v\cdot\nabla\Psi = \eta_{\mathrm{m}}(\partial_r^2 + 3r^{-1}\partial_r + \partial_z^2)\Psi$, the heat equation in $\mathbb{R}^5$ for a function of (r, z), with a bounded drift on $[t_0, t]$. So $|\Psi| \le \|\Psi(t_0)\|_{L^\infty}$ by the maximum principle for bounded solutions. The Lorentz force is $-r\Psi^2 e_r$ plus a gradient. It has $f_\theta = 0$ and is bounded near the axis, and $(\operatorname{curl} f)_\theta = -r\partial_z(\Psi^2)$, so the rescaled source (4.8a) is $-(-t_n)^{2+2h}\partial_Z(\Psi_n^2)$, where Ψ_n is rescaled like θ_n, and tends to zero in $W^{-1,\infty}_{\mathrm{loc}}$; step (iii) applies with Ψ in place of θ, except that in $\mathbb{R}^5$ the drift term $v\cdot\nabla\Psi$ is not in divergence form, so the bounds on Ψ in $C^{1/2}$ and in $C^{1,1/2}$ in x come instead from interior L^p estimates for the heat equation in $\mathbb{R}^5$ with a bounded drift. □

Remark 1 (General magnetic fields and zero scalar diffusivity)**:** *For a general magnetic field bounded near the singular point, once $u = v$, interior parabolic estimates on cylinders of radius $c(-t)^{1/2+h}$ give $|(B\cdot\nabla)B| \le C(-t)^{-1/2-h}$ and $|\operatorname{curl}((B\cdot\nabla)B)| \le C(-t)^{-1-2h}$; the circulation source is then integrable in time since $h < 1/2$, and the rescaled sources of CIV's Section 4 tend to zero. A local Caccioppoli estimate for the induction equation, using $u \in L^{10/3}$, gives $\int_{Q_r}|(B\cdot\nabla)B|^2 \le Cr^2$ on parabolic cylinders Q_r of radius r, which places $(B\cdot\nabla)B$ in the parabolic Morrey space used by Gustafson, Kang and Tsai and in $L^2_tL^2_x$, and controls the enstrophy forcing. For general magnetohydrodynamics the remaining input is $u = v$, which Theorem 1 gives whenever* $\operatorname{curl}((B\cdot\nabla)B)$ *satisfies (UC), for example when $B(\cdot, t)$ is real-analytic. With globally axisymmetric data on $\mathbb{R}^3$, Proposition 3 also holds at $\kappa = 0$: the solution stays axisymmetric, so step (ii) is not needed, and on the regular annulus the vorticity equation in divergence form, with sources $u\otimes\omega - \omega\otimes u$ and the bounded force, gives bounded and then Hölder vorticity by iteration on nested cylinders, hence bounded ∇u, which replaces the parabolic estimates for θ.*

Remark 2 (Electrolytes)**:** *In a homogeneous electrolyte with constant conductivity σ and permeability μ_0, driven by imposed steady electrode currents and fixed magnets in the low magnetic Reynolds number approximation, the imposed current is $J_0 = -\sigma\nabla\phi_0$ with ϕ_0 harmonic in the fluid, and* $\operatorname{curl} B = \mu_0 J_0$, $\nabla\cdot B = 0$, $\operatorname{curl} J_0 = 0$ *give $\Delta B = 0$. So $J_0\times B$ is real-analytic in the fluid interior and satisfies (UC). The induced current $\sigma(-\nabla\phi_1 + u\times B)$ depends on u, so the full system is inductionless magnetohydrodynamics, a closed system, not (1) with a prescribed force.*

Table 3 collects the cases.

Table 3 Force classes and singularities of the CIV class, that is, singularities satisfying (1.7) and (1.8).

Force	**Result**	**Source**
gradient, including uniform gravity	excluded	Theorem 1, case (c)
Coriolis about the symmetry axis	excluded, with (1.5)	Proposition 2
$\operatorname{curl} f$ real-analytic at each t	excluded, with (1.5)	Theorem 1, case (a)
$\operatorname{curl} f$ quasianalytic at each t	excluded, with (1.5)	Theorem 1, case (b)
$\operatorname{curl} f$ axisymmetric	excluded, with (1.5)	Theorem 1, case (c)
Boussinesq buoyancy on $\mathbb{T}^3$	excluded, θ bounded	Proposition 3
toroidal magnetohydrodynamics on $\mathbb{R}^3$	excluded, B_θ/r bounded	Proposition 3
OpenAI force	not excluded	[12, Remark 1.2]

5 The Gevrey class of the OpenAI force

A function on a ball is in the Gevrey class $s > 1$ with constants C, A if $|\partial_x^\alpha f| \le C A^{|\alpha|} |\alpha|!^s$ for all α. We use the notation of the pulse construction of [2]: the band index ℓ, $Q_\ell = 2^{-\ell}$, $\varepsilon = Q_\ell^h$, and the pulse amplitude parameter $S_* = \ell^2$ [2, (6.1)]. A wave label γ of band ℓ has the angular harmonic kp_γ with $k = \lceil \varepsilon^{-1/2} \rceil$ [2, (6.27)]; along a pulse, parametrized by $v \in [0, L_s]$ with $L_s \asymp S_*$, the envelope is $P(v)$ [2, (6.27), (7.16)], the seed cutoff is ψ and the seed amplitude is t_m [2, (7.40)], and χ is the stage cutoff of [2, Lemma 5.4]. A choice of the stage cutoffs (a_j) is *admissible* if it satisfies [2, Lemma 5.4] and the following additional condition: for each sufficiently large band there is a seed point at which the full label cutoff is bounded below and $|\psi'| \ge c/L_s$, and at which the sum of the absolute values of the correction-stage contributions to the residual coefficient at the seed harmonic is at most half the primary seed coefficient.

Proposition 4 (Gevrey obstruction for admissible cutoffs)**:** *Let y_0 be the singular point.*

(a) If $f(\cdot, t)$ satisfies Gevrey-s bounds on a fixed ball around y_0 with C and A independent of t, then $|f(x,t)| \le C \exp(-c\, d^{-1/(s-1)})$, where $d \le (2X_{\mathrm{ext}} q)^{1/2}$ is the radial distance from x to the region $X \ge X_{\mathrm{ext}}$.

(b) For an admissible choice of the (a_j), for each large ℓ there is a point y_ℓ with $q \asymp Q_\ell$ and $|f(y_\ell)| \ge \exp(-C\ell^2)$.

(c) For such a choice, f lies in no spatial Gevrey class $s > 1$ with constants uniform in t, and in no Denjoy–Carleman class with $M_k = k! \exp(\iota k^p)$, $\iota > 0$, $0 < p < 2$, with constants uniform in t.

(d) A version of the construction with $S_* = Q_\ell^{-\varrho}$ in place of ℓ^2, whose seed survives as in (b), has a force in a Gevrey class s with constants uniform in t only if $s \ge 1 + 1/(2\varrho)$. If it keeps the choice $k = \lceil \varepsilon^{-1/2} \rceil$ of [2, (6.27)] and the step of the proof of [2, Lemma 7.1] that imposes $S_*^2(\varepsilon + \varepsilon^2 + k^{-1}) \le 1$, then $\varrho \le h/4$, and (6) gives $s \ge 1 + 2/h > 1.19 \times 10^5$.

Proof (a) The force and all its derivatives vanish at y_0 and on the region $X \ge X_{\mathrm{ext}}$ near y_0 [2, (3.4), Theorem 3.1(iii)]; the final cutoffs equal one near the origin. Taylor's formula of order n along the radial segment of length d from x to that region gives $|f(x,t)| \le C A^n n!^{s-1} d^n$, and the choice $n \asymp (Ad)^{-1/(s-1)}$ gives the bound.

(b) Fix a label γ of band ℓ and the seed point of the admissibility condition. Such points exist: choose an interior sub-box on which the full label cutoff has a positive lower bound; for large ℓ its image under the fast torus coordinate contains a fundamental domain, since the fast oscillation grows exponentially in ℓ while the sub-box shrinks by at most a power of ℓ, so it meets a seed window with $|\psi'| \ge c/L_s$. Project the cylindrical components of f onto the angular harmonic kp_γ of the label at that (r, z, t). (i) The seed term $\psi'(v) t_m$ of [2, (7.40)] is kept unchanged in the residual, and its coefficient is at least $c\varepsilon^{1/2} S_*^{1/4} L_s^{-1} P(v)$ by [2, (7.21), (7.25)]. (ii) Mean fields have no such harmonic, and waves of other labels vanish there because their supports are disjoint [2, Lemma 6.1]. (iii) Every other term of the same label is smaller by a factor $\varepsilon^{0.18}$ times a power of S_*, which tends to zero: the seed has amplitude order $\varepsilon^{1/2}$ and the other wave terms order $\varepsilon^{0.68}$ or higher [2, (6.29), (9.9)], all with the same weight $P(v)$ of [2, (6.29)], so that the comparison holds pointwise in v, products with mean fields keep the envelope of the wave, the retained tails of each stage obey the bounds of [2, (9.8)], and the stage cutoff $\chi(a_j q)$ depends only on (z, t), so it commutes with the fast and radial derivatives, and since $a_{j+1} \ge 2a_j$ at most one stage is in transition at a point. (iv) Finitely many stages are nonzero at the point, and at the selected seed point admissibility makes the sum of their absolute values at most half the seed coefficient. (v) $P(v) \ge \exp(-C S_*)$ [2, (7.16)], and a Fourier coefficient is at most the maximum on the circle. With $S_* = \ell^2$ this gives (b).

(c) The point y_ℓ lies at radial distance at most $C Q_\ell^{1/2}$ from the region $X \ge X_{\mathrm{ext}}$, so (a) gives $|f(y_\ell)| \le C \exp(-c\, 2^{\ell/(2(s-1))})$, which is smaller than $\exp(-C\ell^2)$ for large ℓ, contradicting (b). For $1 < p < 2$ the same comparison applies to $M_k = k! \exp(\iota k^p)$, whose Taylor bound decays like $\exp(-c\,(\log 1/d)^{p/(p-1)})$ with $p/(p-1) > 2$. For $0 < p \le 1$, $\exp(\iota k^p) \le \exp(\iota k)$, so the class is real-analytic, and a force that vanishes on an open subset of the connected ball is zero on the ball by the identity theorem, contradicting (b).

(d) With $S_* = Q_\ell^{-\varrho}$, step (v) gives a seed of size $\exp(-c Q_\ell^{-\varrho})$ at distance $C Q_\ell^{1/2}$ from the region where f vanishes, and (a) bounds it by $\exp(-c Q_\ell^{-1/(2(s-1))})$; so $\varrho \ge 1/(2(s-1))$. The condition of Lemma 7.1 with $\varepsilon = Q_\ell^h$ requires $S_*^2 \varepsilon \le 1$, hence $\varrho \le h/2$. That proof takes $k = \lceil \varepsilon^{-1/2} \rceil \le 2\varepsilon^{-1/2}$ [2, (6.27)], so $S_*^2 k^{-1} \ge Q_\ell^{h/2 - 2\varrho}/2$, and the condition fails for all large ℓ unless $\varrho \le h/4$. □

Admissible choices exist. Let $u^{[j]}$ be the velocity after stage j, before the summation cutoffs are applied. The stage increments $u^{[j]} - u^{[j-1]}$ are constructed before the cutoffs are chosen [2, (9.15), Proposition 9.9], and the derivatives of $\chi(a_j q)$ obey bounds with constants independent of a_j [2, Lemma 5.4]: q depends only on z and t, and a time derivative of the cutoff costs a factor q^{-1} and an axial derivative a factor $q^{-1/2+h}$ [2, (5.28)], each of which at $q \asymp Q_\ell$ is at most a constant times the factor that the same derivative of an amplitude coefficient costs [2, (9.19)]. So step (iii) of the proof of (b) gives numbers K_j and m_j, independent of the cutoffs, such that at a seed point of band ℓ constructed as in (b), where $q \asymp Q_\ell$, the contribution of stage j at the seed harmonic is at most $K_j \ell^{m_j} Q_\ell^{0.18h}$ times the primary seed coefficient. At that point $\ell \le C \log(1/q)$, so after K_j is enlarged the requirement $K_j \log(1/q)^{m_j} q^{0.18h} \le 2^{-j-1}$ for $0 < q \le a_j^{-1}$ has the form of [2, (5.40)]; it may be added at step j,

and it holds once a_j is large. Stage j vanishes where $q \geq a_j^{-1}$, so at these seed points the correction stages sum to at most half the primary seed coefficient. OpenAI's construction does not impose this requirement: the only citation of (5.40) adds no requirement of this form, and the bound $q^{g_j/2}$ of OpenAI's (5.37), where $g_j = hj/10$ [2, proof of Proposition 9.9], is far larger than $\exp(-C\ell^2)$. So (b) and (c) are proved for the admissible members of the family, and are open for the others. Classes whose constants grow as $t \to 1$ are not excluded; Theorem 1 allows such classes.

Remark 3 (A conditional Gevrey floor)**:** *A floor that does not retain the sufficient step of Lemma 7.1 holds for every construction of this type under three premises read from the text of [2]: (P1) each pulse stays in one band, so $\varepsilon L_s \leq C$; (P2) the pulse growth rate per unit of the pulse parameter v is bounded by a constant independent of the band, as follows from the base shear bound and the perturbation energy identity [2, (7.22)]; (P3) the seed enters the force as in (b), with its peak bounded below by a power of Q_ℓ. Then the ratio of peak to seed is at most* $\exp(CQ_\ell^{-h})$*, the seed is at least* $\exp(-CQ_\ell^{-h})$ *up to powers of Q_ℓ, and the argument of (d) with $\varrho = h$ gives $s \geq 1 + 1/(2h) > 2.99 \times 10^4$. Premise (P1) is inferred from the statements that the slow time moves at rate $-\varepsilon$ along a pulse and that a band spans a bounded range of slow time; we have not traced it through the paper's Proposition 9.1.*

6 How far the exact core axisymmetry can be weakened

The results of Section 4 use (1.8), the exact vanishing of w on a core. CIV note that their proof "uses the exact vanishing in (1.8), and that we do not know how to replace it by an algebraic rate of smallness". We show what a quantitative replacement costs. A qualitative replacement gains nothing: under (1.6), $u(\cdot, t)$ is real-analytic on $B(1)$ at each fixed t [12, Corollary 2.2], so if $w(\cdot, t)$ vanishes to infinite order at $x = 0$, then $u - R_\varphi u$ is analytic and vanishes to infinite order at 0, since the rotations fix the axis, and the identity theorem gives $w(\cdot, t) = 0$ on $B(1)$.

Lemma 1 (Propagation across an interval)**:** *Let $\vartheta = \log 3/\log 12$. Let g be smooth on $[-2\varsigma, 2\varsigma]$ with $|g^{(m)}| \leq M\, m!\, \delta^{-m}$ there for all $m \geq 0$, and $|g| \leq \epsilon \leq M$, $M > 0$, on $[-\varsigma, \varsigma]$, where $\varsigma \leq \delta/12$. Then* $\sup_{[-2\varsigma,2\varsigma]} |g| \leq 3M(\epsilon/M)^\vartheta$.

Proof If $\epsilon = M$ the bound with $m = 0$ suffices. If $\epsilon = 0$, the bound holds for every $\epsilon' \in (0, M)$ in place of ϵ, and letting $\epsilon' \to 0$ gives it for $\epsilon = 0$. Otherwise let $n = \lceil \log(M/\epsilon)/\log 12 \rceil \geq 1$ and let p be the Taylor polynomial of g of degree $n - 1$ at 0. The remainder satisfies $|g(x) - p(x)| \leq M(|x|/\delta)^n$. On $[-\varsigma, \varsigma]$ this is at most $M12^{-n} \leq \epsilon$, so $|p| \leq 2\epsilon$ there. By Chebyshev's inequality for polynomials, on $[-2\varsigma, 2\varsigma]$

$$\begin{aligned} |p| &\leq 2\epsilon\, T_{n-1}(2) \\ &\leq 2\epsilon\, 4^{n-1} \\ &< 2\epsilon\, (M/\epsilon)^{\log 4/\log 12} \\ &= 2M(\epsilon/M)^\vartheta, \end{aligned}$$

where the strict inequality uses $n - 1 < \log(M/\epsilon)/\log 12$. On $[-2\varsigma, 2\varsigma]$ the remainder is at most $M6^{-n} \leq M(\epsilon/M)^{\log 6/\log 12} \leq M(\epsilon/M)^\vartheta$. □

Proposition 5 (Quantitative criterion)**:** *Let $0 < R_1 < R_2 \leq 1$ and $M, d, a > 0$, and fix t. Suppose that $|(e \cdot \nabla)^m w(\cdot, t)| \leq M\, m!\, d^{-m}$ on $B(R_2)$ for every unit vector e and every $m \geq 0$, and that $|w(\cdot, t)| \leq \epsilon \leq M$ on $B(a)$. Put $d_* = \min(d, R_1/6)$, $\varsigma = \min(a/2, d_*/12, (R_2 - R_1)/2)$ and $N = \lfloor R_1/\varsigma \rfloor + 1$. Then*

$$\sup_{B(R_1)} |w| \leq 7.2\, M(\epsilon/M)^{\vartheta^N}, \tag{14}$$

and

$$\|w(\cdot, t)\|_{C^3(B(R_1/2))} \leq C\, M\, d_*^{-3} (\epsilon/M)^{\vartheta^N/4}. \tag{15}$$

If (1.5) and (1.7) hold and the right side of (15) is bounded uniformly in t, then $(0, 0)$ is a regular point.

Proof Put $\rho_k = a + k\varsigma$ and $\varpi_0 = \epsilon/M$, and suppose $|w| \leq M\varpi_k$ on $B(\rho_k)$. For x with $\rho_k \leq |x| < \rho_k + \varsigma$, put $e = x/|x|$ and $c = x - 2\varsigma e$; then $|c| + \varsigma \leq \rho_k$, so $|w| \leq M\varpi_k$ on the segment $c + [-\varsigma, \varsigma]e$, and the segment $c + [-2\varsigma, 2\varsigma]e$ lies in $B(R_2)$ while $\rho_k \leq R_1$, since $\varsigma \leq (R_2 - R_1)/2$. Lemma 1, applied with $\delta = d$ to $\langle w(c + ve), \xi \rangle$ as a function of v, for each unit vector ξ, gives $|w| \leq M\varpi_{k+1}$ on $B(\rho_{k+1})$, with $\varpi_{k+1} = \min(1, 3\varpi_k^\vartheta)$. By induction $\varpi_k \leq 3^{1/(1-\vartheta)} \varpi_0^{\vartheta^k}$, and $3^{1/(1-\vartheta)} < 7.2$. The induction runs while $\rho_k \leq R_1$, so it stops after at most N steps, since $\rho_N > R_1$. Since $\varpi_0 \leq 1$ and

$\vartheta^k \geq \vartheta^N$ for $k \leq N$, the bound at the last step is at most $7.2(\epsilon/M)^{\vartheta^N}$, which gives (14). If $\epsilon = 0$, (14) gives $w = 0$ on $B(R_1)$, so (15) follows. For $\epsilon > 0$, write $\beta = 7.2(\epsilon/M)^{\vartheta^N}$. If $\beta > 1$, the hypothesis with $m \leq 3$ and polarization give (15) directly. If $\beta \leq 1$, use along each direction e the third difference with step $d_*\beta^{1/4}$ at four nodes in $B(R_1)$, which needs $3d_*\beta^{1/4} \leq R_1/2$ and holds since $d_* \leq R_1/6$: its value is at most $8Md_*^{-3}\beta^{1/4}$, and its error is at most $CMd_*^{-3}\beta^{1/4}$ by the bound on the fourth derivative. Polarization recovers the mixed derivatives, and lower orders are bounded in the same way. The last statement is CIV's Remark 2.4, which derives regularity from a uniform C^3 bound on w, after rescaling $B(R_1/2)$ to $B(1)$. □

When M, $1/d$ and $1/a$ grow like powers of $1/\tau$, N grows like a power of $1/\tau$, and the criterion needs $\epsilon/M \leq \exp(-\exp(K\tau^{-b}))$ for some $b, K > 0$. When d and a are bounded below, algebraic smallness suffices. The analytic bounds are hypotheses here, not consequences of (1.5) to (1.7): the shear flow $u = g(x_1, t)e_3$ with $g = d^6 x_1/(x_1^2 + d^2)$ and $d = e^{-1/\tau}$ solves (1) with the force $(\partial_t g - \partial_{x_1}^2 g)e_3$, which satisfies (1.5) and (1.6); it has $\mathsf{P}u = 0$ because g is odd in x_1, and its analyticity radius d tends to zero faster than any power of τ.

Proposition 6 (Sharpness of the double exponential)**:** *For $0 < d < 1$ let $G_d(z) = \exp(-\exp(\pi(1/4 - z)/(2d)))$ and $w_d = \big(G_d(x_1) - \mathsf{P}G_d\big)e_3$, where $\mathsf{P}G_d$ is the angular mean of $G_d(x_1)$. Then w_d is divergence-free with $\mathsf{P}w_d = 0$, it extends holomorphically to the tube $|\operatorname{Im} x| < d$ with $|w_d| < 2$ there, $|w_d| \leq 2\exp(-\exp(\pi/(16d)))$ on $B(1/8)$, and $w_d(1/2, 0, 0) \to \frac{2}{3}e_3$ as $d \to 0$.*

Proof The field $c(x_1, x_2)e_3$ has zero divergence, and $\mathsf{P}(G_d(x_1)e_3) = (\mathsf{P}G_d)e_3$ because e_3 is fixed by the rotations, so $\mathsf{P}w_d = 0$. For $|\operatorname{Im} z| < d$ the argument of $\exp(\pi(1/4 - z)/(2d))$ lies in $(-\pi/2, \pi/2)$, so its real part is positive and $|G_d(z)| < 1$; the rotated arguments $x_1 \cos\varphi - x_2 \sin\varphi$ have imaginary part below d as well. On $B(1/8)$, $1/4 - x_1 \geq 1/8$ for every rotated point, which gives the second bound. At $(1/2, 0, 0)$, $G_d(1/2) \to 1$, and $G_d(\frac{1}{2}\cos\varphi)$ tends to 1 for $|\varphi| < \pi/3$ and to 0 for $\pi/3 < |\varphi| \leq \pi$, so $\mathsf{P}G_d \to 1/3$. □

By Cauchy's estimates w_d satisfies the hypotheses of Proposition 5 with M bounded and analytic radius comparable to d, and its smallness on the core $B(1/8)$ is double-exponential in $1/d$, yet $|w_d|$ is close to $2/3$ at distance $1/2$. So no criterion that uses only the analytic bound, the analytic radius and the core radius can accept a smallness weaker than $\exp(-\exp(c/d))$, and the threshold of Proposition 5 is sharp up to the constant in the exponent. This limits the method; w_d is not a solution of (1).

7 Conclusion

The OpenAI construction proves what it states, and this paper does not dispute it. What the construction needs from its force cannot come from the physical sources examined here. Its own constants make the matching radius of its leading field more than 7.20×10^{47} times its core length, so in water a leading field whose core is not smaller than a molecule has a matching radius more than 10^{12} times the Hubble radius. Its singularity is of the CIV class, and no gradient force, and no force bounded in C^2 up to the blowup time that is a Coriolis force about the symmetry axis or has real-analytic, quasianalytic or axisymmetric curl, can produce a singularity of that class; nor can Boussinesq buoyancy on $\mathbb{T}^3$ or the toroidal Lorentz force on $\mathbb{R}^3$. For an admissible choice of its stage cutoffs, its force is in no Gevrey class with constants uniform in time.

Four problems remain open.

1. A closed coupled system, Boussinesq or magnetohydrodynamics with general fields, that produces a singularity of the CIV class, or a proof that none does.
2. A replacement of (1.8) by an algebraic rate of smallness under assumptions that do not include the analytic bounds of Proposition 5; by Proposition 6 it needs more structure than those bounds.
3. A Gevrey version of the construction and its least index, or a proof that none exists.
4. A proof or disproof of premise (P1) for the construction; if (P1) holds, the floor $s > 2.99 \times 10^4$ of Remark 3 holds under the remaining premises (P2) and (P3).

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Data availability. No datasets were generated or analyzed in this work. The physical constants used are taken from the cited sources [8–10].

Code availability. A Python script that recomputes every number stated in this paper, together with negative controls that it must reject, is provided as supplementary material.

Use of language models. Language models (Anthropic Claude, OpenAI GPT and Google Gemini) were used in drafting and reviewing this paper, in checking its citations against the cited sources, and in writing the script that recomputes its numbers. The author takes full responsibility for the content.

Author contributions. The author is the sole author of this work.

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